

		is removed.
14	C	From $pV = nRT$, when p and V are tripled, the temperature would be $9T$ New temperature = $(200 + 273) \times 9 = 4257 \text{ K}$ $= 4257 - 273 = 3984 = 4000 \text{ }^{\circ}\text{C}$ From $\frac{1}{2} m \langle c^2 \rangle = \frac{1}{2} m v^2 = \frac{3}{2} kT \rightarrow v^2 = \frac{3kT}{m} \rightarrow v = \sqrt{\frac{3kT}{m}}$, when the new temperature is $9T$, the new r.m.s. speed = $3v$
15	D	Option A: The kinetic energy of the oscillator is proportional to square of the